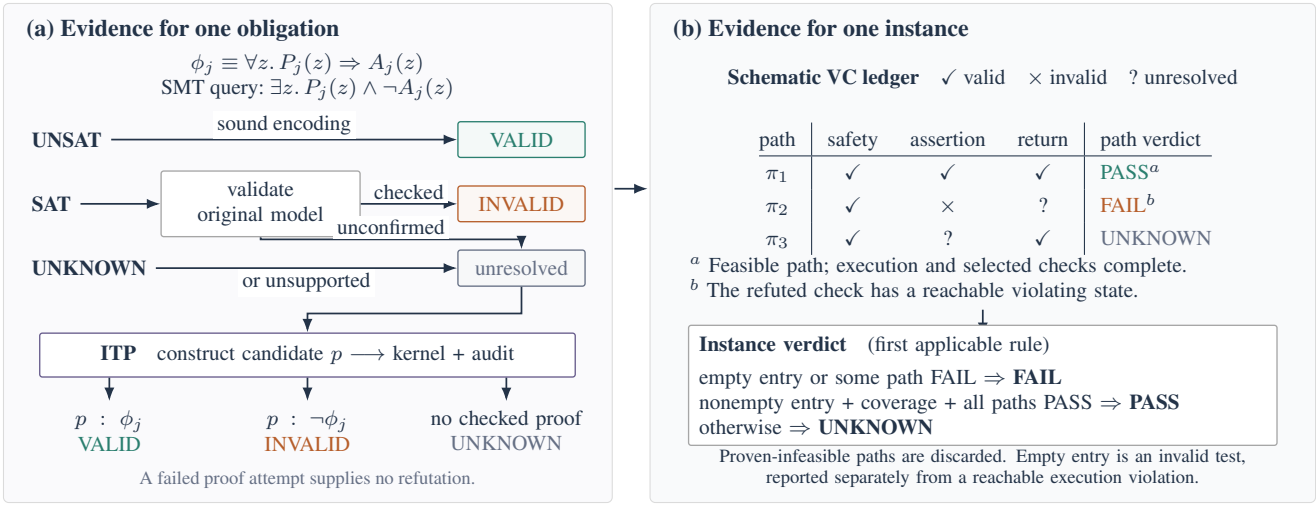

(a) Evidence for one obligation
$\phi_j \equiv \forall z.\, P_j(z) \Rightarrow A_j(z)$
SMT query: $\exists z.\, P_j(z) \wedge \neg A_j(z)$
UNSAT
sound encoding
VALID
SAT
validate original model
checked
INVALID
unconfirmed
UNKNOWN
or unsupported
unresolved
ITP construct candidate $p \longrightarrow$ kernel + audit
$p : \phi_j$
VALID
$p : \neg\phi_j$
INVALID
no checked proof
UNKNOWN
A failed proof attempt supplies no refutation.
(b) Evidence for one instance
Schematic VC ledger ✓ valid × invalid ? unresolved
path | safety | assertion | return | path verdict
$\pi_1$ | ✓ | ✓ | ✓ | PASS[a]
$\pi_2$ | ✓ | × | ? | FAIL[b]
$\pi_3$ | ✓ | ? | ✓ | UNKNOWN
[a] Feasible path; execution and selected checks complete.
[b] The refuted check has a reachable violating state.
Instance verdict (first applicable rule)
empty entry or some path FAIL ⇒ FAIL
nonempty entry + coverage + all paths PASS ⇒ PASS
otherwise ⇒ UNKNOWN
Proven-infeasible paths are discarded. Empty entry is an invalid test, reported separately from a reachable execution violation.